

Review retrospective:
the differentiation-under-the-integral base, $G'_n = G_{n+1}$

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Abstract

A soundness review of `Laplace/OneD/AnharmonicPartitionHigherDeriv.lean`: the differentiation-under-the-integral step that powers the anharmonic cumulant ladder — $\partial_h \int (-(tx))^n e^{-t(L+hx)}|_0 = \int (-(tx))^{n+1} e^{-tL}$ — plus its $n = 1, 2$ specialisations. Result: **a clean fidelity pass with no edits**. The dominated-differentiation proof is correct end to end, and both opens are load-bearing.

1 Setting

This is the analytic engine below the cumulant algebra reviewed in `AnharmonicFourthCumulant`: where that file differentiates ratios G_1/G_0 symbolically, this one proves the underlying fact that each moment integral $G_n(h) = \int (-(tx))^n e^{-t(L+hx)}$ has h -derivative G_{n+1} , by differentiating under the integral sign. It is the integral-side counterpart to the pure-algebra cumulant files, and — as expected — it is the one that actually needs `MeasureTheory`.

2 What was reviewed

The general step `anharmonic_partition_deriv_step` and its two specialisations `anharmonic_partition_secondDeriv` ($n = 1$, giving $\int (tx)^2 e^{-tL}$) and `anharmonic_partition_thirdDeriv` ($n = 2$, giving $-\int (tx)^3 e^{-tL}$).

3 Fidelity / soundness findings

Sound; checked end to end. The h -derivative of $(-(tx))^n e^{-t(L+hx)}$ is $(-(tx))^{n+1} e^{-t(L+hx)}$ (the constant-in- h prefactor times the chain-rule factor $-tx$), matching the stated RHS at $h = 0$. The dominator $t^{n+1} e^{t/(2c)} |x|^{n+1} e^{-(t/2)L}$ is the product of $(t|x|)^n$ and the imported single-factor perturbation bound, integrable via `integrable_abs_pow_mul_exp_neg_t_anharmonic (n+1)` at $t/2$; it dominates $\|F'(h, x)\|$ for $|h| < 1$. All seven data of `hasDerivAt_integral_of_dominated_loc_of_deriv_le` are correctly supplied (the pointwise bound and pointwise `HasDerivAt` are in fact proved everywhere, then packaged AE). The $n = 1, 2$ instances correctly use $(-(tx))^2 = (tx)^2$ and $(-(tx))^3 = -(tx)^3$. The reviewer confirmed all of this.

One report-only note: the corollary docstrings title themselves “second/third h -derivative of Z ”, while the theorems literally give the h -derivative-at-0 of the first/second-derivative *integral expression*. This is accurate in context — the body names “the first-derivative integral”, and the

$n = 0$ step identifies that integral with Z' — so the title is shorthand for a true chain, not an overclaim; left as is.

4 Simplifications applied

None — the bucket was empty. Both open MeasureTheory (eighteen bare $\int$, plus AESTronglyMeasurable/Integrable/volume) and open Topology (bare $\mathcal{N}$, $\forall^f\text{-in-}\mathcal{N}$) are load-bearing; there is no open Real (all Real.* qualified); the single import supplies the coercivity, integrability, and pointwise bound/derivative lemmas; and the dense DCT proof has no dead local or no-op step. No worktree.

5 Disagreements and open questions

None.

6 Lessons

- **The integral-side file is exactly where MeasureTheory earns its keep.** The fourth-cumulant review predicted the higher cumulant files inherit a dead open MeasureTheory; that holds for the *algebra* files but is reversed here — the differentiation-under-the-integral base has eighteen bare $\int$. The measure-vs-algebra split is per-file, and the bare- $\int$ count settles it instantly.
- **Stronger-than-needed pointwise hypotheses are not dead code.** The proof establishes the bound and the derivative pointwise everywhere and then packages them as AE statements for the DCT lemma. That is not redundancy to trim — it is the cleanest way to feed the AE-shaped Mathlib API.

7 Follow-ups

- The remaining anharmonic arc: AnharmonicPartitionDerivGeneralH, AnharmonicGibbsRegularity, AnharmonicGibbsObservableMonomials, AnharmonicKappa3Affine; and the harmonic regularity/observable files.